DAS732: Data Visualization / Term 1 (2026-27) / Programming Assignment 2 (A2)
International Institute of Information Technology Bangalore

Announcement Date: Sep 30, 2026 (Wed)
Last Date for Dataset Selection: Oct 04, 2026 (Sun)
Submission Deadline: 11:59 pm IST, Nov 01, 2026 (Sun)

Summary: Familiarization with Scivis and Infovis methods

A2 tasks and requirements for a group assignment:

- **Scivis**: Use Python to implement SciVis methods for three tasks: (i) color mapping, (ii) contour mapping, and (iii) quiver plots (optionally, experiment with streamlines).

  Generate animation GIFs using 5-10 time instances (i.e., at least 5 and at most 10 temporal samples) of a scalar field.

- **Infovis**:
  i. Use Gephi or NetworkX to implement a node-link diagram of a sufficiently complex network as a task, and compare at least 3 graph layout algorithms with user interactions.
  ii. Use JavaScript to implement Infovis methods as browser visualizations with user interactions for two tasks: (i) the parallel coordinates plot with edge bundling, and (ii) zoomable treemap visualizations.
  iii. Use Python libraries to generate time-series visualizations of horizon plots and ridgeline plots. If the dataset does not have time-series data, use group-wise distributions.

Description of the Scivis dataset:

Data source: https://www.climatologylab.org/gridmet.html

The gridMET dataset provides daily meteorological data for a period of over 40 years. Each team will work on a contiguous three-month period (e.g., January-March 2020, or November 2019-January 2020) that the team has already selected. The time instances of the dataset can be sampled either uniformly or randomly to select 5-10 samples. The rationale behind choosing the time period and the time instances must be explained in the assignment report.

Selection of the dataset slice for each team:

1. The TAs will provide clear instructions on how to select a slice for each team through a Google Form.
2. The selection will follow a first-come, first-served policy.

Steps for Downloading the gridMET dataset:

1. On the gridMET page (given as the data source above) in a browser, find the "Download" tab in the navigation menu at the top and click on it.

2. Scroll down the page, and under the Direct Downloads section, find the link labeled "Direct download of NetCDF files".
3. Click on this link to be redirected to the repository that contains the NetCDF files.
4. You should now see a list of .nc files. The files are organized by variable and year in the format: <variable abbreviation>_<year>.nc.
5. Use vs (Wind speed at 10 m) and th (Wind direction at 10 m) for the Quiver plot. Use other variables for contour mapping and heatmap or color mapping.

Infovis tasks:

1. Generate appropriate networks, multivariate, and hierarchical datasets from the dataset given in A1.
2. If network tasks are difficult to implement with the A1 dataset, email the instructor about the need for network data sufficiently before the deadline.

A2 assignment specifications:

1. Assessment is based on the report, images folder, and codebase, as applicable.
    a. The report shall have images with appropriate captions, and the inferences in the text shall be written about these images.
    b. The images in the folder may be named using their figure indices in the report so that the high-resolution images are available for assessment. It is sometimes hard to see the relatively lower-resolution images included in the report.
    c. All visualizations should be complete with text information for title, legend, axes, etc.
    d. The screenshots from Gephi and other GUIs shall be complete and of high resolution.
    e. The codebase shall include a README for implementation.
    f. A video demo is not required for this assignment. Hence, it is important to include sufficient inferences in the report.
    g. The report shall be professionally prepared as done for RWA. Inclusion of appropriate references, as possible, is encouraged.
2. There are 3 tasks for each of the visualization domains, i.e., Scivis and Infovis. Each student shall implement at least one Scivis and one Infovis visualization task.
    a. The combined outcome of the team can be proportional to the team's strength.
    b. The report shall clearly state the contributions of each team member.
    c. Apart from data cleaning and processing (as required), there cannot be joint ownership of the tasks. If a team violates this rule, marks will be deducted accordingly.
    d. No two team members will do the same task.
3. The contour mapping shall be done using the Marching Squares algorithm and/or contour fill algorithm. Explain the rationale behind the choice in the report.
4. The color mapping shall include experimentation with different colormaps, e.g., continuous, discrete, logarithmic, etc., and different color palettes. As appropriate, you could comment about the use of sequential and diverging colormaps.
5. The color mapping shall comment on parametric mapping using global vs local maxima and minima, across different timesteps.
6. The quiver plot can experiment with one of the following:
    a. Grid sampling experiments along with a comparison of magnitude-proportional and same-sized vectors.

    b. Compare and contrast the streamline method and quiver plots, and consider experiments with different seeding strategies for the streamline implementation.
7. Node-link diagram experiments shall include different graph layouts, and optionally, any desired node filtering implementation.
8. Parallel coordinate plot implementation is complete only with user interactions. Hence, either axis reordering or brushing or both shall be implemented.
9. Treemap experiments shall include different remodels of tabular data to the tree data structure. Optionally, different spatial partitioning strategies can be experimented with.
10. Provide at least two different variations with different groups or temporal granularity (year vs month), likewise for group-wise data, for each of the two visualizations, i.e., horizon plots and ridgeline plots.

Rubrics for assessment:

The total points are out of 30, which will be scaled to 20 in the final grade.

- Completeness - 4 points
    - All experiments as mentioned in the assignment description must be attempted.
    - The report, code, images, README, etc. must be complete.
- Correctness - 8 points
    - Inferences must be correct with respect to the context of the assigned dataset, its inherent nature (e.g., space and time), etc.
- Quality of Inferences - 8 points
    - The complexity of arriving at the inferences, effective use of visualizations, etc., will be considered.
- Data Handling - 10 points
    - Preprocessing steps, any other data-specific transformations, etc., will be checked.